COMPLETE INTERSECTION LOCALIZES AT PRIMES

THE STACKS PROJECT AUTHORS

Editorial note. The proof marks its reduction step '(this is the key step of the proof)'; retained verbatim.

Allowed standard prerequisites (editorial).

- Avramov proposition. proposition-avramov (same chapter, named). Reduces from the local pair (A_p, (A^^)_q) to the complete ring.
- Regular quotient criterion by regular sequence. lemma-quotient-regular-ring-by-regular-sequence (same chapter, named). Certifies the completed localization is CI from the explicit presentation.

Source reference. Original reference 00MC: focused-seeds/geometry/S9-dpa-ci-good/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09Q0: sources/stacks-dpa/pinned-original-source.tex, source lines 1979–1979. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 09Q2: sources/stacks-dpa/pinned-original-source.tex, source lines 2059–2059. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a Noetherian local ring. Let $\mathfrak{p} \subset A$ be a prime ideal. If A is a complete intersection, then $A_\mathfrak{p}$ is a complete intersection too.*

Proof. Choose a prime $\mathfrak{q}$ of $A^\wedge$ lying over $\mathfrak{p}$ (this is possible as $A \to A^\wedge$ is faithfully flat by Algebra, Lemma 00MC). Then $A_\mathfrak{p} \to (A^\wedge)_\mathfrak{q}$ is a flat local ring homomorphism. Thus by Proposition 09Q2 we see that $A_\mathfrak{p}$ is a complete intersection if and only if $(A^\wedge)_\mathfrak{q}$ is a complete intersection. Thus it suffices to prove the lemma in case A is complete (this is the key step of the proof).

Assume A is complete. By definition we may write $A = R/(f_1, \ldots, f_r)$ for some regular sequence $f_1, \ldots, f_r$ in a regular local ring R. Let $\mathfrak{q} \subset R$ be the prime corresponding to $\mathfrak{p}$. Observe that $f_1, \ldots, f_r \in \mathfrak{q}$ and that $A_\mathfrak{p} = R_\mathfrak{q}/(f_1, \ldots, f_r)R_\mathfrak{q}$. Hence $A_\mathfrak{p}$ is a complete intersection by Lemma 09Q0. □

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